University of Dhaka

Assignment - 3

Course Title: Advanced Econometrics-II
Course Code: 408

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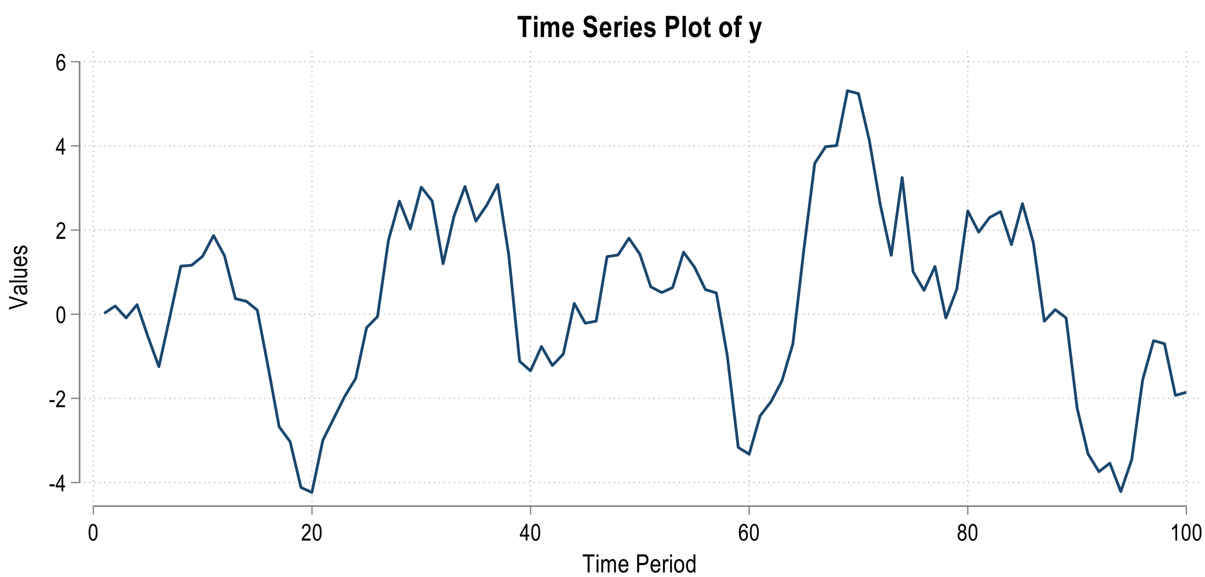
Question:

Attached is a STATA data file with a time series variable "y". First, using the Box-Jenkins methodology, determine which ARMA model this variable follows. After estimating the ARMA model, please forecast 10 steps ahead of "y" with a 95% confidence interval. Extra marks for better visual representation.

Solution:

Following the Box-Jenkins Methodology, we will follow (a) Identification (b) Estimation (c) Diagnostic Test, these 3 procedures to find the best Univariate ARMA model that this variable y follows.

(a) <u>Identification:</u>

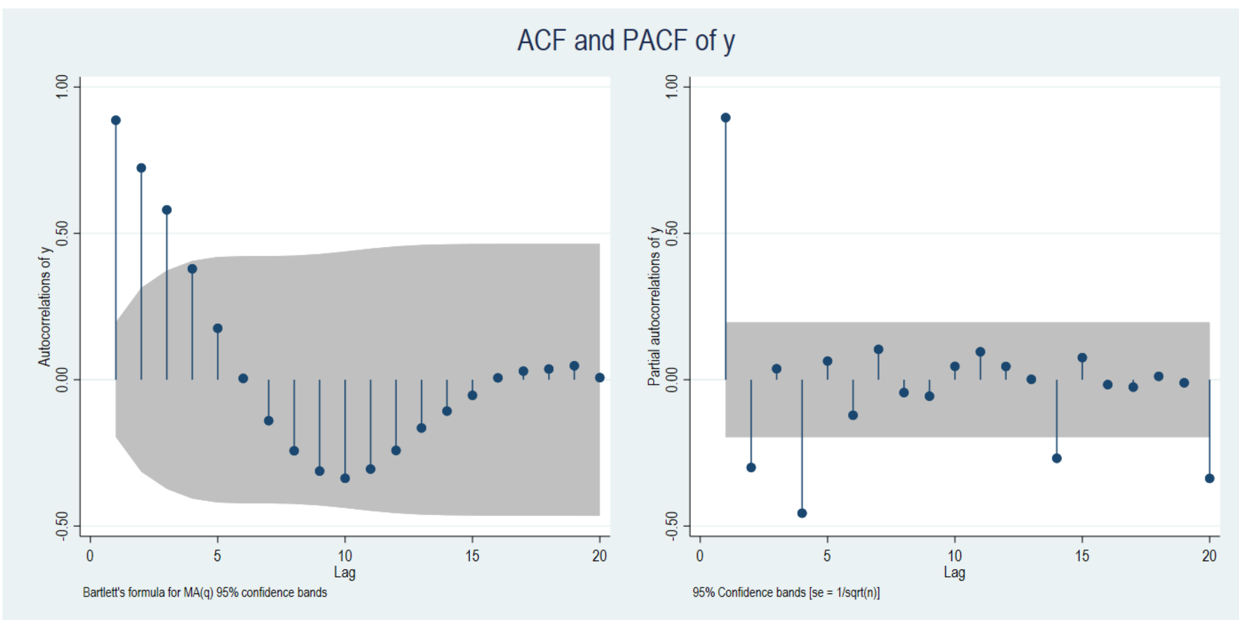


H0: Random walk without drift, d = 0

	Test Statistics	Dicky Fuller Critical Values 1%	5%	10%
Z(t)	-3.622	-3.517	-2.894	-2.582

MacKinnon approximate p-value for Z(t) = 0.0054

The Time Series plot of the variable y clearly shows no significant trend. And the Augmented Dicky-Fuller Test also rejects the Null Hypothesis of a Unit Root Process at lag 4. So, we can conclude that variable y is stationary.



Both the Autocorrelation and Partial Auto-correlation Function of y variable show significant spikes till lag 3 and exponential decay. So, I assume the presence of ARMA (2,3), ARMA (3,2), ARMA (3,3), ARMA (3,4), ARMA (4,3). Now I will estimate all the assumed models and compare their information criteria to get the parsimonious model.

Information Criteria:

Model	*N*	*LL (model)*	*df*	*AIC*	*BIC*
ARMA (2,3)	100	-124.3935	7	262.7871	281.0233
ARMA (3,2)	100	-125.8966	7	265.7933	284.0295
ARMA (3,3)	100	-124.1635	8	264.327	285.1684
ARMA (3,4)	100	-123.6252	9	265.2504	288.6969
ARMA (4,3)	100	-123.5017	9	265.0035	288.45

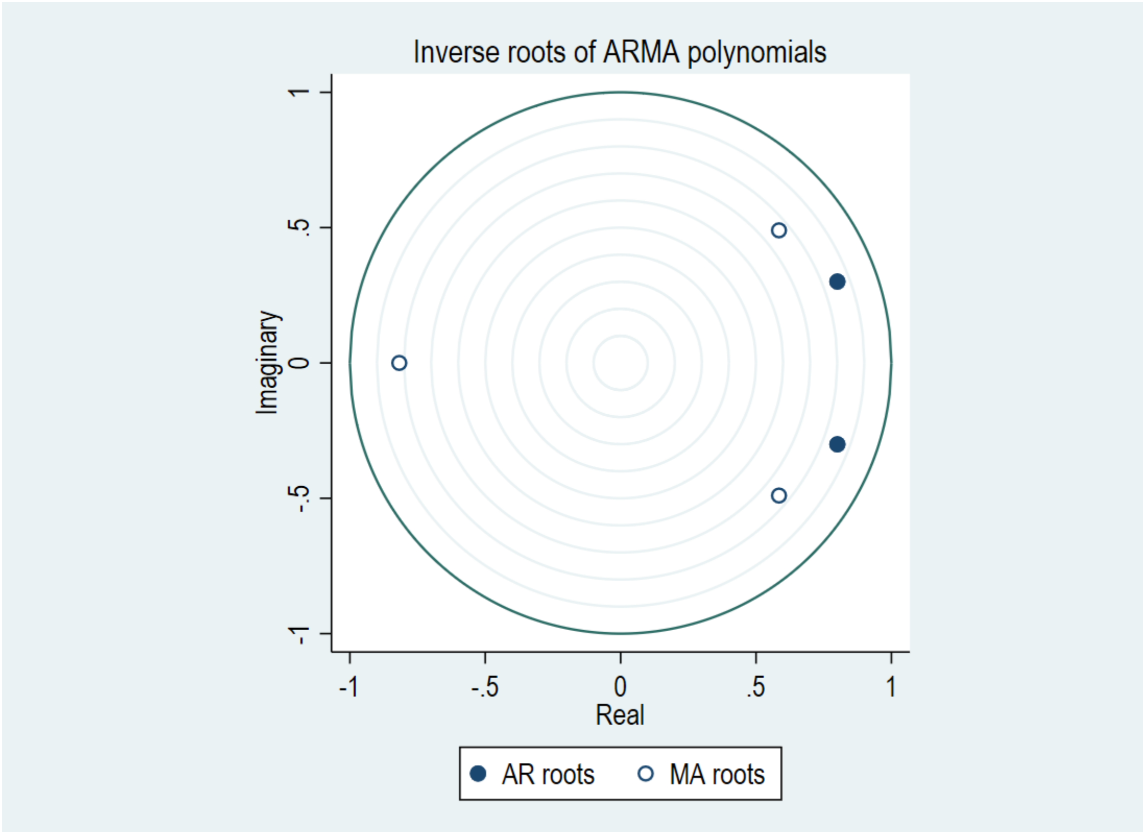
Among those guessed model, ARMA (2,3) yields the lowest AIC and BIC so this is my chosen Univariate model for the variable y.

(b) *Estimation:*

ARIMA (2,3) regression

y	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
Constant	.219	.51	0.43	.668	-.781	1.218	
ar							
L	1.601	.146	10.94	0	1.314	1.888	***
L2	-.731	.12	-6.12	0	-.966	-.497	***
ma							
L	-.352	.145	-2.43	.015	-.636	-.068	**
L2	-.374	.127	-2.95	.003	-.623	-.126	***
L3	.476	.116	4.11	0	.249	.702	***
Constant	.828	.057	14.52	0	.716	.94	***
Mean dependent var		0.275	SD dependent var			2.162	
Number of obs		100	Chi-square			867.399	
Prob > chi2		.	Akaike crit. (AIC)			262.787	

**** p<.01, ** p<.05, * p<.1*



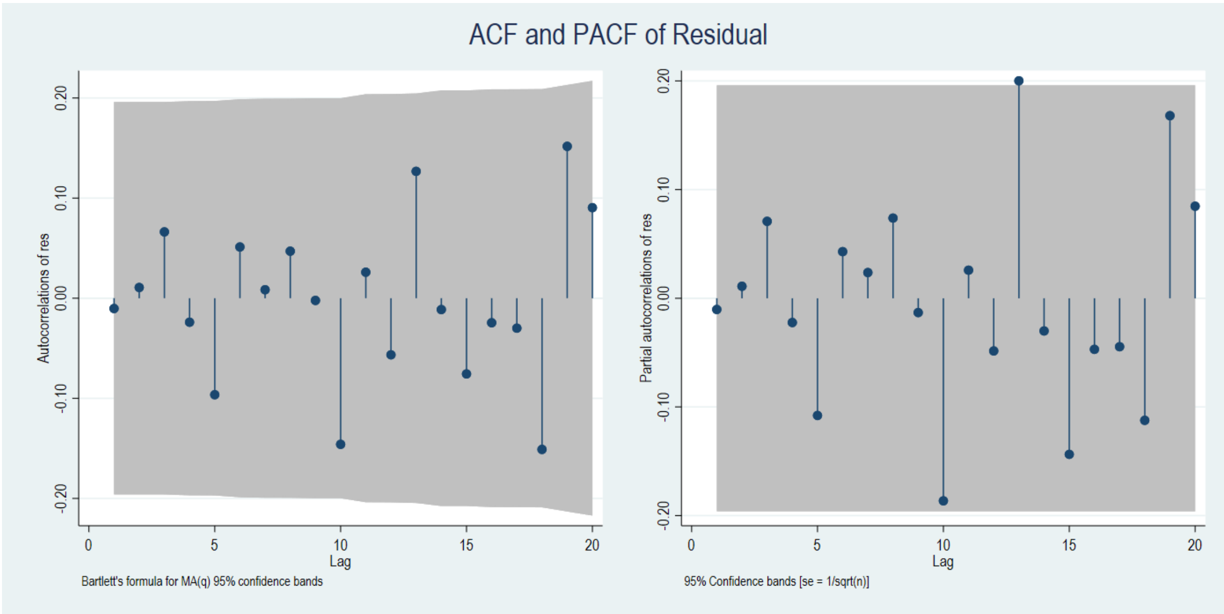
After estimating the ARMA (2,3) model we can see all the lag coefficients of AR and MA process are statistically significant at 1% confidence intervals. And the roots diagram shows us that all the roots of AR and MA process are within the unit circle. So, we can say ARMA (2,3) model is acceptable for estimation.

(c) <u>Diagnostic Test:</u>

Portmanteau test for white noise

Portmanteau (Q) statistics	=	33.0943
Prob > chi2(40)	=	0.7721

The Ljung-Box test or Q test at lag 40 do not reject the Null Hypothesis of the residuals being white noise. The Autocorrelation and Partial Autocorrelation function of the residuals also justify this white noise process for lag 20. This diagnostic test satisfies the residuals being white noise and according to the steps of Box Jenkins Methodology ARMA (2,3) model is accepted as the univariate process of the variable y.



(d) <u>10 Steps Ahead Forecast:</u>